



Code Snippets

1	Documentation	3
1.1	SQLite	3
1.2	Regex	3
2	Code snippets	4
2.1	Label line spacing	4
2.2	Launching another application	4
2.3	Reading mp3 headers	4
2.4	Writing wave file	5
2.5	Initialize GPS	7
2.6	Minimize EditText height	7
2.7	Changing color of selected text in a Label	7
2.8	SelectionEnd in an EditText	8
2.9	Turn off text correction	8
2.10	SQLite BLOB UPDATE	8
2.11	Rename a folder	8
2.12	Webviewscrolling	8
2.13	Change color of Android ProgressBar	9
2.14	Change the padding (margins) of an EditText view	10
2.15	Search on the Internet	10
2.16	Get the pressure on the screen	10
2.17	Calculate the Number of Days Between Two Dates	11
2.18	Display a PDF file	11
2.19	Select Case with True	11
2.20	Fill an array with random numbers without repetition	11
2.21	Change Alpha property of a view	12
2.22	Getting a views' index	12
2.23	How Can I Hide the Scrollbar on Scrollview	12
2.24	How to call the internal calculator	12
2.25	Get pixel colors	13
2.26	Get device type	13
2.27	Generate a Click event	14
2.28	Remove the scrollbar from a ScrollView	14
2.29	Pseudo Custom Controls	15
2.30	Elipsize a Label	16
2.31	Get the dpi values (dot per inch)	16
2.32	Android settings indents	17
3	Classes	19
3.1	Draggable view class	19

Contributors:

Klaus Christl

Forum users

1 Documentation

In this chapter some links that might be interesting.

1.1 SQLite

<http://www.sqlite.org/lang.html>

1.2 Regex

<http://regexlib.com/?AspxAutoDetectCookieSupport=1>

<http://www.regular-expressions.info/>

2 Code snippets

Code snippets gathered from the forum.

2.1 Label line spacing

```
Dim Obj1 As Reflector
Obj1.Target = lbl
Obj1.RunMethod3("setLineSpacing", 1, "java.lang.float", 2, "java.lang.float")
```

2.2 Launching another application

```
Dim in As Intent
Dim pm As PackageManager
in = pm.GetApplicationIntent("com.google.android.youtube")
If in.IsInitialized Then StartActivity(in)
StartActivity(in)
```

2.3 Reading mp3 headers

http://www.mpgedit.org/mpgedit/mpeg_format/mpeghdr.htm#MPEGTAG

```
Sub Activity_Create(FirstTime As Boolean)
    LogMp3Tags(File.DirRootExternal, "1.mp3")
End Sub

Sub LogMp3Tags(Dir As String, FileName As String)
    Dim raf As RandomAccessFile
    raf.Initialize2(Dir, FileName, True, True)
    If raf.Size < 128 Then
        Log("No TAG found.")
        Return
    End If
    Dim buffer(128) As Byte
    raf.ReadBytes(buffer, 0, 3, raf.Size - 128)
    If BytesToString(buffer, 0, 3, "UTF8") <> "TAG" Then
        Log("No TAG found.")
        Return
    End If
    'Title
    raf.ReadBytes(buffer, 0, 30, raf.CurrentPosition)
    Log("Title=" & ConvertBytesToString(buffer, 30))
    'Artist
    raf.ReadBytes(buffer, 0, 30, raf.CurrentPosition)
    Log("Artist=" & ConvertBytesToString(buffer, 30))
    'Album
    raf.ReadBytes(buffer, 0, 30, raf.CurrentPosition)
    Log("Album=" & ConvertBytesToString(buffer, 30))
    'Year
    raf.ReadBytes(buffer, 0, 4, raf.CurrentPosition)
    Log("Album=" & ConvertBytesToString(buffer, 4))
    'Comment
```

```

raf.ReadBytes(buffer, 0, 30, raf.CurrentPosition)
Log("Comment=" & ConvertBytesToString(buffer, 30))
'Genre
Dim genre As Int
genre = raf.ReadUnsignedByte(raf.CurrentPosition)
Log("Genre=" & genre)
End Sub

Sub ConvertBytesToString(Buffer() As Byte, MaxLength As Int) As String
    For i = 0 To MaxLength - 1
        If Buffer(i) = 0 Then Return BytesToString(Buffer, 0, i, "UTF8")
    Next
    Return BytesToString(Buffer, 0, MaxLength, "UTF8")
End Sub

```

2.4 Writing wave file

```

Sub Globals
...
    Type TWaveHeader _
        (ChunkID(4) As Byte, _
        ChunkSize As Int, _
        Format(4) As Byte, _
        Subchunk1ID(4) As Byte, _
        Subchunk1Size As Int, _
        AudioFormat As Short, _
        NumChannels As Short, _
        SampleRate As Int, _
        ByteRate As Int, _
        BlockAlign As Short, _
        BitsPerSample As Short, _
        Subchunk2ID(4) As Byte, _
        Subchunk2Size As Int)

    Dim WavHead As TWaveHeader
    Dim NumSamples As Int: NumSamples = 1000
    Dim WavData(2*NumSamples) As Short
End Sub

Sub Activity_Create(FirstTime As Boolean)
    Dim l1 As Long
    Dim Buf(44+4*NumSamples), Buf2(4*NumSamples) As Byte
    Dim bc As ByteConverter
    Dim RAB, RAF As RandomAccessFile

...

    WavHead.Initialize
    bc.LittleEndian = True
    RAB.Initialize3(Buf, True)

    WavHead.ChunkID = "RIFF".GetBytes("UTF8")
    WavHead.Format = "WAVE".GetBytes("UTF8")
    WavHead.Subchunk1ID = "fmt ".GetBytes("UTF8")
    WavHead.Subchunk2ID = "data".GetBytes("UTF8")

    WavHead.Subchunk1Size = 16
    WavHead.AudioFormat = 1

```

```

    WavHead.NumChannels = 2
    WavHead.SampleRate = 44100
    WavHead.BitsPerSample = 16
    WavHead.ByteRate = WavHead.SampleRate * WavHead.NumChannels * WavHead.BitsPerSample / 8
    WavHead.BlockAlign = WavHead.NumChannels * WavHead.BitsPerSample / 8
    WavHead.Subchunk2Size = NumSamples * WavHead.NumChannels * WavHead.BitsPerSample / 8
    WavHead.ChunkSize = 36 + WavHead.Subchunk2Size

    l1 = GenSin(WavData, 0, 32000, 5000, NumSamples, 44100)
    Buf2 = bc.ShortsToBytes(WavData)

    RAB.WriteByte(WavHead.ChunkID, 0, 4, 0)
    RAB.WriteInt(WavHead.Subchunk1Size, 4)
    RAB.WriteByte(WavHead.Format, 0, 4, 8)
    RAB.WriteByte(WavHead.Subchunk1ID, 0, 4, 12)
    RAB.WriteInt(WavHead.Subchunk1Size, 16)
    RAB.WriteShort(WavHead.AudioFormat, 20)
    RAB.WriteShort(WavHead.NumChannels, 22)
    RAB.WriteInt(WavHead.SampleRate, 24)
    RAB.WriteInt(WavHead.ByteRate, 28)
    RAB.WriteShort(WavHead.BlockAlign, 32)
    RAB.WriteShort(WavHead.BitsPerSample, 34)
    RAB.WriteByte(WavHead.Subchunk2ID, 0, 4, 36)
    RAB.WriteInt(WavHead.Subchunk2Size, 40)

    RAB.WriteByte(Buf2, 0, 2*l1, 44)

    RAF.Initialize2(File.DirDefaultExternal, "data.wav", False, True)
    RAF.WriteByte(buf, 0, 44+2*l1, 0)

    ...
End Sub

' GenSin = (2 * pi * t * freq) / samplerate [return sample count]
' buf - sample buffer
' ch - channel 0-left, 1-right
' v - volume (magnitude multiplier)
' fr - frequency
' ns - number of samples max
' sr - sample rate [clock time = ( 1.0 / (ns/ sr) )]

Sub GenSin(buf() As Short, ch As Int, v As Float, fr As Float, ns As Int, sr As Int) As Int
    Dim i, j As Int
    Dim g, h, tp, fsr As Float

    tp = 2 * cPI
    fsr = sr

    For i=0 To ns-1
        j = 2 * i + ch
        g = (i * tp * fr) / fsr
        h = v * Sin(g)
        buf(j) = h
    Next

    Return(i)
End Sub

```

2.5 Initialize GPS

```
If GPS1.GPSEnabled = False Then StartActivity (GPS1.LocationSettingsIntent)
```

2.6 Minimize EditText height

Based on Reflection library.

```
Sub Globals
    Dim txt1 As EditText
End Sub

Sub Activity_Create(FirstTime As Boolean)
    txt1.Initialize("txt1")
    Dim c As Canvas
    Dim b As Bitmap
    b.InitializeMutable(1, 10)
    c.Initialize2(b)
    txt1.Text = "Hello"

    Dim size As Float
    size = c.MeasureStringHeight(txt1.Text, txt1.Typeface, txt1.TextSize)
    txt1.Gravity = Bit.Or(Gravity.TOP, Gravity.LEFT) 'Set gravity to top le
ft
    Activity.AddView(txt1, 10dip, 10dip, 100dip, size + 6dip)

    txt1.Color = Colors.White 'Removes the view edges
    Dim r As Reflector
    r.Target = txt1
    'Set padding to minimum
    r.RunMethod4("setPadding", Array As Object(0, 1dip, 0, 0), _
        Array As String("java.lang.int", "java.lang.int", "java.lang.int",
"java.lang.int"))
End Sub
```

2.7 Changing color of selected text in a Label

```
Sub Activity_Create(FirstTime As Boolean)
    Dim l As Label
    l.Initialize("l")
    Activity.AddView(l, 0, 0, 100dip, 100dip)
    l.Text = "Some text"
    SetColorList(l, Colors.Yellow, Colors.Blue)
End Sub

Sub l_Click

End Sub

Sub SetColorList (Label1 As Label, DefaultColor As Int, PressedColor As Int)
    Dim sd As StateListDrawable
    Dim clrs(2) As Int
    Dim states(2,1) As Int
    clrs(0) = DefaultColor
    clrs(1) = PressedColor
```

```

states(0, 0) = sd.State_Pressed
Dim r As Reflector
Dim csl As Object
csl = r.CreateObject2("android.content.res.ColorStateList", _
    Array As Object(states, clr), Array As String("[I", "[I"))
r.Target = Label1
r.RunMethod4("setTextColor", Array As Object(csl), _
    Array As String("android.content.res.ColorStateList"))
End Sub

```

2.8 SelectionEnd in an EditText

Based on Reflection library.

```

Sub GetSelectionEnd(txt As EditText) As Int
    Dim r As Reflector
    r.Target = txt
    Return r.RunMethod("getSelectionEnd")
End Sub

```

2.9 Turn off text correction

```
EditText1.InputType = 52428
```

2.10 SQLite BLOB UPDATE

```
SQL1.ExecNonQuery2("INSERT INTO table2 VALUES('smiley', ?)", Array As
Object(Buffer))
```

2.11 Rename a folder

```

Sub Activity_Create(FirstTime As Boolean)
    RenameFolder(File.DirRootExternal, "test1", "test2")
End Sub

Sub RenameFolder(Parent As String, CurrentFolder As String, NewFolder)
    Dim p As Phone
    Dim args(2) As String
    args(0) = File.Combine(Parent, CurrentFolder)
    args(1) = File.Combine(Parent, NewFolder)
    p.Shell("mv", args, Null, Null)
End Sub

```

2.12 Webviewscrolling

So, in summary, to scroll a WebView to a particular DOM element, write a JavaScript function to do the scrolling:

```
function scrollToElement(id) {
    var elem = document.getElementById(id);
    var x = 0;
    var y = 0;

    while (elem != null) {
        x += elem.offsetLeft;
        y += elem.offsetTop;
        elem = elem.offsetParent;
    }
    window.scrollTo(x, y);
}
```

and then from your Android app (Java code), tell your WebView to load a URL:

```
webView.loadUrl("javascript:scrollToElement('" + elemId + "')");
```

There are some issues with this approach, such as the scroll will not be nicely animated, but the general mechanism works.

2.13 Change color of Android ProgressBar

Based on Reflection library.

```
Sub Globals
    Dim pb As ProgressBar
End Sub
Sub Activity_Create(FirstTime As Boolean)
    pb.Initialize("pb")
    Dim gd As GradientDrawable
    gd.Initialize("TOP_BOTTOM", Array As Int(Colors.Blue, Colors.Red))
    gd.CornerRadius = 3dip
    SetProgressDrawable(pb, gd)

    pb.Progress = 50
    Activity.AddView(pb, 10dip, 10dip, 300dip, 50dip)
End Sub

Sub SetProgressDrawable(p As ProgressBar, drawable As Object)
    Dim r As Reflector
    Dim clipDrawable As Object
    clipDrawable =
r.CreateObject2("android.graphics.drawable.ClipDrawable", _
                Array As Object(drawable, Gravity.LEFT, 1), _
                Array As String("android.graphics.drawable.Drawable",
"java.lang.int", "java.lang.int"))
    r.Target = p
    r.Target = r.RunMethod("getProgressDrawable") 'Gets the
layerDrawable
    r.RunMethod4("setDrawableByLayerId", _
```

```
        Array As Object(r.GetStaticField("android.R$id",  
"progress"), clipDrawable), _  
        Array As String("java.lang.int",  
"android.graphics.drawable.Drawable"))  
End Sub
```

2.14 Change the padding (margins) of an EditText view

```
Dim refl As Reflector  
refl.Target=EditFeld  
refl.RunMethod4("setPadding", Array As Object(Left, Top, Right, Bottom), _  
        Array As String("java.lang.int", "java.lang.int", _  
"java.lang.int", "java.lang.int"))
```

2.15 Search on the Internet

```
Dim SearchWord As String  
SearchWord = "apples"
```

```
Dim i As Intent  
i.Initialize(i.ACTION_VIEW, "http://www.google.com/#q=" & SearchWord)
```

2.16 Get the pressure on the screen

```
Sub Activity_Create(FirstTime As Boolean)  
  
    Dim r As Reflector  
    Dim btn As Button  
    btn.Initialize("")  
    Activity.AddView(btn, 10dip, 10dip, 300dip, 300dip)  
    r.Target = btn  
    r.SetOnTouchListener("btn_touch")  
End Sub  
  
Sub btn_touch (viewtag As Object, action As Int, X As Float, Y As Float,  
motionevent As Object) As Boolean  
    Dim r As Reflector  
    r.Target = motionevent  
    Dim pressure As Float  
    pressure = r.RunMethod("getPressure")  
    Log(pressure)  
    Return True  
End Sub
```

2.17 Calculate the Number of Days Between Two Dates

```
Sub Activity_Create(FirstTime As Boolean)
    DateTime.SetTimeZone(0)
    Dim Date1, Date2 As Long
    DateTime.DateFormat = "M/d/yyyy h:mm:ss a Z"
    Date1 = DateTime.Parse("12/29/2011 11:04:17 AM -5")
    Date2 = DateTime.Parse("12/27/2011 11:04:17 AM -5")
    Log(DaysBetweenDates(Date1, Date2))
End Sub

Sub DaysBetweenDates(Date1 As Long, Date2 As Long)
    Return Floor((Date2 - Date1) / DateTime.TicksPerDay)
End Sub
```

2.18 Display a PDF file

```
Sub Btn_Tarif_Click
    Dim nom_file As String
    nom_file = "file:///sdcard/_lelong/fiches_tarif/" & nomdoc
    'Msgbox(nom_file,"nom fichier")
Dim zz As Intent 'Requires a reference to the Phone library
zz.Initialize(zz.ACTION_VIEW, nom_file)
zz.SetType("application/pdf")
'zz.WrapAsIntentChooser("Choose PDF Viewer")
StartActivity(zz)
End Sub
```

2.19 Select Case with True

```
i = 10
Select True
Case (i < 9)
    Log("False")
Case (i = 10)
    Log("True")
End Select
```

2.20 Fill an array with random numbers without repetition

```
Sub Activity_Create (FirstTime As Boolean)
    Dim numbers(10) As Int
    'put numbers 1 - 10 in the array
    For i = 0 To 9
        numbers(i) = i + 1
    Next
    ShuffleArray(numbers)
    For i = 0 To 9
        Log(numbers(i)) 'print the numbers to the log
    Next
End Sub

Sub ShuffleArray(arr() As Int)
    For i = arr.Length - 1 To 0 Step -1
```

```
        Dim j, k As Int
        j = Rnd(0, i + 1)
        k = arr(j)
        arr(j) = arr(i)
        arr(i) = k
    Next
End Sub
```

2.21 Change Alpha property of a view

```
Sub SetAlpha(Control As View, Alpha As Int)
    Dim r as Reflector
    r.Target = Control.Background
    r.RunMethod2("setAlpha", Alpha, "java.lang.int")
End Sub
```

2.22 Getting a views' index

```
Sub Activity_Create(FirstTime As Boolean)
    Activity.LoadLayout("1")
    Log(Panels.IndexOf(Panel1, EditText2))
End Sub

Sub Panels.IndexOf(p As Panel, v As View) As Int
    For i = 0 To p.NumberOfViews - 1
        If p.GetView(i) = v Then Return i
    Next
    Return -1
End Sub
```

2.23 How Can I Hide the Scrollbar on Scrollview

```
Dim r As Reflector
r.Target = ScrollView1
r.RunMethod2("setVerticalScrollBarEnabled", False, "java.lang.boolean")
```

2.24 How to call the internal calculator

```
Sub Calculator
    Dim i As Intent
    i.Initialize("", "")
    i.SetComponent("com.android.calculator2/.Calculator")
    Try
        StartActivity(i)
    Catch
        ToastMessageShow("Calculator app not found.", True)
    End Try
End Sub
```

2.25 Get pixel colors

```
Sub Activity_Create(FirstTime As Boolean)
    Dim argb() As Int
    argb = GetARGB(Colors.Transparent)
    Log("A = " & argb(0))
    Log("R = " & argb(1))
    Log("G = " & argb(2))
    Log("B = " & argb(3))
End Sub

Sub GetARGB(Color As Int) As Int()
    Dim res(4) As Int
    res(0) = Bit.UnsignedShiftRight(Bit.And(Color, 0xff000000), 24)
    res(1) = Bit.UnsignedShiftRight(Bit.And(Color, 0xff0000), 16)
    res(2) = Bit.UnsignedShiftRight(Bit.And(Color, 0xff00), 8)
    res(3) = Bit.And(Color, 0xff)
    Return res
End Sub
```

2.26 Get device type

```
Sub Activity_Create(FirstTime As Boolean)
    If GetDevicePhysicalSize > 6 Then
        '7'' or 10'' tablet
    Else
        'phone
    End If
End Sub

Sub GetDevicePhysicalSize As Float
    Dim lv As LayoutValues
    lv = GetDeviceLayoutValues
    Return Sqrt(Power(lv.Height / lv.Scale / 160, 2) + Power(lv.Width / lv.
Scale / 160, 2))
End Sub
```

2.27 Generate a Click event

```
Sub Globals
    Dim sp As Spinner
End Sub

Sub Activity_Create(FirstTime As Boolean)
    sp.Initialize("sp")
    sp.AddAll(Array As String("a", "b", "c", "d"))
    Activity.AddView(sp, 10dip, 10dip, 200dip, 50dip)
End Sub

Sub Activity_Click
    OpenSpinner(sp)
End Sub

Sub OpenSpinner(s As Spinner)
    Dim r As Reflector
    r.Target = s
    r.RunMethod("performClick")
End Sub
```

2.28 Remove the scrollbar from a ScrollView

The code needs the Reflection library.

```
Dim r As Reflector
r.Target = ScrollView1
r.RunMethod2("setVerticalScrollBarEnabled", False, "java.lang.boolean")
```

2.29 Pseudo Custom Controls

Here is a very simplistic implementation:

Code module named C1

```
Sub Process_Globals
    Type MyView (p As Panel, b1 As Button, l1 As Label)
End Sub

Sub CreateMyView(ButtonEvent As String, ButtonText As String, LabelText As String) As MyView
    Dim m As MyView
    m.Initialize
    m.p.Initialize("")
    m.b1.Initialize(ButtonEvent)
    m.b1.Text = ButtonText
    m.l1.Initialize("")
    m.l1.Text = LabelText
    m.b1.Tag = m 'store the structure in the button's Tags property.
    m.p.AddView(m.b1, 10dip, 10dip, 100dip, 40dip)
    m.p.AddView(m.l1, 10dip, 50dip, 100dip, 40dip)
    Return m
End Sub

Sub ChangeValue(m As MyView, value As String)
    m.l1.Text = value
End Sub
```

Activity:

```
Sub Process_Globals
End Sub

Sub Globals
    Dim m1, m2 As MyView
End Sub

Sub Activity_Create(FirstTime As Boolean)
    m1 = c1.CreateMyView("m1", "Button #1", "Label #1")
    Activity.AddView(m1.p, 10dip, 10dip, 100dip, 200dip)
    m2 = c1.CreateMyView("m2", "Button #2", "Label #2")
    Activity.AddView(m2.p, 10dip, 110dip, 100dip, 200dip)
End Sub

Sub m1_Click
    c1.ChangeValue(m1, "new value")
End Sub

Sub m2_Click
    c1.ChangeValue(m2, "new value2")
End Sub
```

2.30 Elipsize a Label

Needs the reflection library.

```
Sub Activity_Create(FirstTime As Boolean)
    Activity.LoadLayout(1)
    Label1.Text = "this is a very long text."
    SetEllipsize(Label1, "END")
End Sub
Sub SetEllipsize(TextView As Label, Mode As String)
    Dim r As Reflector
    r.Target = TextView
    r.RunMethod2("setLines", 1, "java.lang.int")
    r.RunMethod2("setHorizontallyScrolling", True, "java.lang.boolean")
    r.RunMethod2("setEllipsize", Mode, "android.text.TextUtils$TruncateAt")
End Sub
```

2.31 Get the dpi values (dot per inch)

Needs the reflection library.

```
Dim Xdpi, Ydpi As Float
Dim r As Reflector
r.Target = r.GetContext
r.Target = r.RunMethod("getResources")
r.Target = r.RunMethod("getDisplayMetrics")
Xdpi = r.GetField("xdpi")
Ydpi = r.GetField("ydpi")
```

2.32 Android settings indents

Code :

```
Dim DoIntent As Intent
DoIntent.Initialize("android.settings.LOCATION_SOURCE_SETTINGS", "")
StartActivity(DoIntent)
```

Action Details(Activity action)

ACCESSIBILITY_SETTINGS = Show settings for accessibility modules.

ADD_ACCOUNT = Show add account screen for creating a new account.

AIRPLANE_MODE_SETTINGS = Show settings to allow entering/exiting airplane mode.

APN_SETTINGS = Show settings to allow configuration of APNs.

APPLICATION_DETAILS_SETTINGS = Show screen of details about a particular application.

APPLICATION_DEVELOPMENT_SETTINGS = Show settings to allow configuration of application development-related settings.

APPLICATION_SETTINGS = Show settings to allow configuration of application-related settings.

BLUETOOTH_SETTINGS = Show settings to allow configuration of Bluetooth.

DATA_ROAMING_SETTINGS = Show settings for selection of 2G/3G.

DATE_SETTINGS = Show settings to allow configuration of date and time.

DEVICE_INFO_SETTINGS = Show general device information settings (serial number, software version, phone number, etc.).

DISPLAY_SETTINGS = Show settings to allow configuration of display.

INPUT_METHOD_SETTINGS = Show settings to configure input methods, in particular allowing the user to enable input methods.

INPUT_METHOD_SUBTYPE_SETTINGS = Show settings to enable/disable input method subtypes.

INTERNAL_STORAGE_SETTINGS = Show settings for internal storage.

LOCALE_SETTINGS = Show settings to allow configuration of locale.

LOCATION_SOURCE_SETTINGS = Show settings to allow configuration of current location sources.

MANAGE_ALL_APPLICATIONS_SETTINGS = Show settings to manage all applications.

MANAGE_APPLICATIONS_SETTINGS = Show settings to manage installed applications.

MEMORY_CARD_SETTINGS = Show settings for memory card storage.

NETWORK_OPERATOR_SETTINGS = Show settings for selecting the network operator.

NFCSHARING_SETTINGS = Show NFC Sharing settings.

NFC_SETTINGS = Show NFC settings.

PRIVACY_SETTINGS = Show settings to allow configuration of privacy options.

QUICK_LAUNCH_SETTINGS = Show settings to allow configuration of quick launch shortcuts.

SEARCH_SETTINGS = Show settings for global search.

SECURITY_SETTINGS = Show settings to allow configuration of security and location privacy.

SETTINGS = Show system settings.

SOUND_SETTINGS = Show settings to allow configuration of sound and volume.

SYNC_SETTINGS = Show settings to allow configuration of sync settings.

USER_DICTIONARY_SETTINGS = Show settings to manage the user input dictionary.

WIFI_IP_SETTINGS = Show settings to allow configuration of a static IP address for Wi-Fi.

WIFI_SETTINGS = Show settings to allow configuration of Wi-Fi.

WIRELESS_SETTINGS = Show settings to allow configuration of wireless controls such as Wi-Fi, Bluetooth and Mobile networks.

EXTRA_AUTHORITIES = Limit available options in launched activity based on the given authority.

2.33 Out of memory problem with orientation change

In many cases the error happens when you change the orientation several times. These kinds of memory problems can be resolved by loading the images only once.

```
Sub Process_Globals
    Dim Image1, Image2 As Bitmap
    Dim ImageView1 As ImageView
End Sub

Sub Activity_Create(FirstTime As Boolean)
    If FirstTime Then
        Image1 = LoadBitmap(...)
        Image2 = LoadBitmap(...)
    End If
    ImageView1.Bitmap =
    '
    '
End Sub
```

3 Classes

3.1 Draggable view class

Main module:

```
'Main activity module
Sub process_globals

End Sub

Sub Globals
    Dim Button1 As Button
    Dim Button2 As Button
    Dim EditText1 As EditText
End Sub

Sub Activity_Create(FirstTime As Boolean)
    Activity.LoadLayout("1")
    Dim dv1, dv2, dv3 As DraggableView
    dv1.Initialize(Activity, Button1)
    dv2.Initialize(Activity, Button2)
    dv3.Initialize(Activity, EditText1)
End Sub
```